

EXERCICES D'ANALYSE CORRIGÉS

```
a=2;b=5
print(a*b,b**a,b+b*a)
>>10 25 15
```

```
c=3.2;d=6.5
print(int(c)*int(d),type(c))
>>18 <class 'float'>
```

```
print(True and False,True or False, not False)
>>False True True
```

```
x,y=3,4.2E+1
x+=2
print("Résultat:",y/x,y//x,y%x)
>>Résultat: 8.4 8.0 2.0
```

```
alpha=0b00100110
beta=0x22
print(alpha*beta +5)
>>1297
```

```
d=abs(-5)
print(pow(d,3))
>>125
```

```
print(ord("T"),chr(0x40))
>>84 @
```

```
lettre = input("Saisir une lettre minuscule:")
chiffre = int (input("Saisir un chiffre:"))
print(lettre.upper()+str(chiffre))
>>K3
```

```
prenom='Isabelle'
print(len(prenom),prenom[3],type(prenom))
>>8 b <class 'str'>
```

```
prenom='Marc'
print('m' in prenom,prenom=='Marco',prenom!='Marque')
>>False False True
```

```
notes=(10,12,14,16)
moy=(notes[0]+notes[1]+notes[2]+notes[3])/4
print("Ma moyenne est de:\n",moy)
>>Ma moyenne est de:
>>13.0
```

```
data="TSI SAINT-CRICQ"
s=data.split(" ")
print("Résultat: ",s,"élément de la liste: ",s[0])
>>Résultat: ['TSI', 'SAINT-CRICQ'] élément de la liste: TSI
```

```
p=range(3,10,2)
print(p[0],p[2],p[-1])
>>3 7 9
```

```
print(C'est FINI!)
    print(C'est FINI!)
      ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
a=2
while a < 12:
    print("Valeur intermédiaire de a:",a)
    a=a+3
print("Valeur finale de a:",a)
>>Valeur intermédiaire de a: 2
>>Valeur intermédiaire de a: 5
>>Valeur intermédiaire de a: 8
>>Valeur intermédiaire de a: 11
>>Valeur finale de a: 14
```

```
def inc(x):
    return x+3
def dec(y):
    return y-5
print(inc(dec(4))*dec(inc(4)))
>>4
```

```
moy=(10,12,13,14.5)
coef=(1,2,0.5,4)
sum=coef[0]+coef[1]+coef[2]+coef[3]
generale=(moy[0]*coef[0]+moy[1]*coef[1]+moy[2]*coef[2]+moy[3]*coef[3])/sum
print("Ma moyenne générale est de:",generale,"arrondie à",round(generale,1))
>>Ma moyenne générale est de: 13.133333333333333 arrondie à 13.1
```

```
prenom='Elsa';nom="Martin"
if len(prenom)+len(nom)%2==0:
```

```
code=2
else:
code=1
print("Votre compte:",prenom+"."+nom+str(code))
>>Votre compte: Elsa.Martin1
```

```
for i in range(1,6,2):
    print(i**3)
>>1
>>27
>>125
```

```
for index in range(1,11):
    if index<=2:
        print("glurb!")
    elif index <6:
        print("zut!")
    else: print("glop!")
>>glurb!
>>glurb!
>>zut!
>>zut!
>>zut!
>>glop!
>>glop!
>>glop!
>>glop!
>>glop!
```

```
def polynome(x):
    return x**2-4*x+2
for i in range(3):
    print(polynome(i))
>>2
>>-1
>>-2
```

```
def affiche(y):
    print(y,y//2,y%3)
for n in range (4,7):
    affiche(n)
>>4 2 1
>>5 2 2
>>6 3 0
```

```
A=[1,2,3,4]
B=[2,3,4,5]
n=len(A)
som=0
for i in range(n):
    som=som+A[i]*B[n-1-i]
print(som)
>>30
```

```
def puissance(a,b):
    if a>b:
        s=a**b
    else:
        s=b**a
    return s
print(puissance(3,4),puissance(6,2))
>>64 36
```

```
print("Calcul factorielle n")
def factoriellen(n):
    result = n
    for i in range (n-1):
        result = result*(n-1)
        n=n-1

    return result
```

```
for i in range(4):
    print(factoriellen(i+1))
>>Calcul factorielle n
>>1
>>2
>>6
>>24
```

```
a=4
while a < 20:
    a=a+2
print("Valeur finale de a:",a)
>>Valeur finale de a: 20
```

```
def inc(x):
    return x+5
def dec(y):
    return y-3
print(inc(dec(3))*dec(inc(2)))
>>20
```

```
moy=(10,12,8,10)
coef=(1,2,2,1)
sum=0;gen=0
for i in range(len(coef)):
    sum=sum+coef[i]
print(sum)
for i in range(len(moy)):
    gen=gen + moy[i]*coef[i]
print(gen)
print("Ma moyenne générale est de:",gen/sum)
>>6
>>60
```

>>Ma moyenne générale est de: 10.0

```
def code(x,y):
    if (len(x)+len(y))%2==0:
        return 2
    else:
        return 1
print(code("Elsa","Martin"))
print(code("Lea","Machin"))
>>2
>>1
```

```
for i in range(2,8,2):
    print(i**2)
>>4
>>16
>>36
```

```
def polynome(x):
    return x**2-2*x+3
for i in range(3):
    print(polynome(i))
>>3
>>2
>>3
```

```
def affiche(y):
    print(y,y//2,y%3)
for n in range (2,5):
    affiche(n)
>>2 1 2
>>3 1 0
>>4 2 1
```

```
from math import sin,pi
for angle in range(0,360,45):
    print("Le sinus de",angle,"° =",sin(angle*pi/180))
>>Le sinus de 0 ° = 0.0

>>Le sinus de 45 ° = 0.7071067811865476
>>Le sinus de 90 ° = 1.0
>>Le sinus de 135 ° = 0.7071067811865476
>>Le sinus de 180 ° = 1.2246467991473532e-16
>>Le sinus de 225 ° = -0.7071067811865475
>>Le sinus de 270 ° = -1.0
>>Le sinus de 315 ° = -0.7071067811865477
```

```
import numpy as np
teta = np.arange(0.0, np.pi, np.pi/3)
print(np.cos(teta))
>>[ 1.  0.5 -0.5]
```

```
ndc1=123456
ndc2=456789
def motdepasse(data):
    return int(data*0.001)
print(motdepasse(ndc1),motdepasse(ndc2))
>>123 456
```

```
ndc=[643456,644567,651231,657894,655612,646214]
liste64=[]
liste65=[]
for i in range(len(ndc)):
    if ndc[i]//10000==64:
        liste64.append(ndc[i])
    if ndc[i]//10000==65:
        liste65.append(ndc[i])
print(liste64+liste65)
>>[643456, 644567, 646214, 651231, 657894, 655612]
```

```
metar="2016/10/28 01:00 SYCJ 280100Z 14526KT CAVOK 26/25 Q1010 NOSIG"
def traitement(message):
    champs=message.split(" ")
    data=champs[2]
    return(data)
print("Le code de la station météo est:",traitement(metar))
>>Le code de la station météo est: SYCJ
```

```
liste1=[2,6.5,12.5]
liste2=[1,3,2]
liste3=[]
for j in range(len(liste1)):
    liste3.append(liste1[j]+liste2[j])
print(liste3)
>>[3, 9.5, 14.5]
```

```
trame="$GPRMC,071139.988,A,4639.8232,N,00021.6810,E,8.95,184.14,141014,1.1,W,A*65"
champs=trame.split(",")
checksum=champs[-1].split("*")
print("checksum =",int(checksum[1]))
>>checksum = 65
```

```
A=[1,2,3,2]
B=[1,3,2,0]
```

```
sum=1
for i in range(len(A)):
    sum=sum*(A[i]+B[-1-i])
print(sum)
>>72
```

```
def puissance(a,b):
    if a>b:
        s=a**b
    else:
        s=b**a
    return s
print(puissance(3,2),puissance(5,2))
>>9 25
```

```
mail='damien.traille@sfr.fr'
data=mail.split('@')
x=data[0].split('.')
print("Nom:",x[1].capitalize()," Prénom:",x[0].capitalize())
>>Nom: Traille Prénom: Damien
```